BIOS 794: Unsupervised Learning: Clustering

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Clustering

- The goal is to identify *natural groupings* in the data without relying on an outcome variable.
- Clustering is an unsupervised learning technique that is one of the most natural exploratory data analysis (EDA) methods.
- Examples in public health:
  - Grouping patients by symptom patterns or biomarker profiles.
  - Identifying communities with similar socioeconomic or health characteristics.
  - Categorizing hospitals based on performance indicators.

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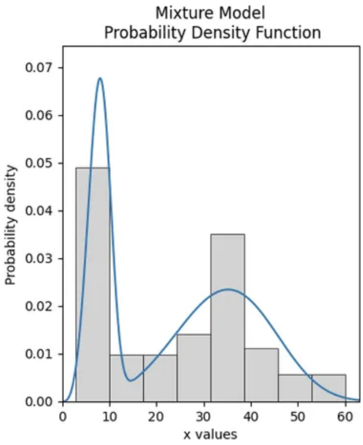
  - Categorizing hospitals based on performance indicators.
- **Differences from classification**
  - we don't know the number of classes
  - we don't know anybody's class (no ground truth)
  - the goal is more exploratory and less predictive
  - we can cluster observations (i.e., the Y_i's), the predictor variables (i.e., the $\boldsymbol{X}_i$'s), or both.

Definition

- Defining what a cluster is can be difficult and seems subjective.
- Statistically, we think of clusters as a mixture distribution:

$$p(\boldsymbol{X}|\Phi) = \sum_{k=1}^{K} I(\boldsymbol{X} \in \mathcal{C}_k) p_k(\boldsymbol{X}|\phi_k).$$

- For example, the $f_k(\boldsymbol{X}|\phi_k)$ could be, for example, different multivariate normal distributions.



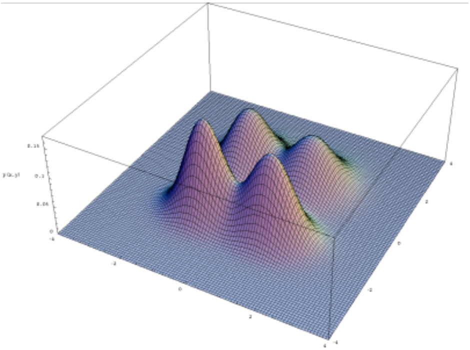


Figure: Multivariate Mixture of Gaussian distributions.

Dissimilarity

- Many clustering algorithms rely on the dissimilarity (or distance) metric used.
- Let's think of $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ as the biomarker profiles (e.g., immune response measures) between patients i and j.
- The dissimilarity between $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ has the following properties
    1. $d(\boldsymbol{x}_i, \boldsymbol{x}_j) \geq 0$
    2. $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = 0 \leftrightarrow \boldsymbol{x}_i = \boldsymbol{x}_j$
    3. $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = d(\boldsymbol{x}_j, \boldsymbol{x}_i)$

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- **Examples**
  - Euclidean: $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = \sqrt{\sum_{k=1}^{p} (x_{ik} - x_{jk})^2}$
  - Manhattan: $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = \sum_{k=1}^{p} |x_{ik} - x_{jk}|$
  - 1−correlation: $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = 1 - \rho_{ij}$

K-means: Concept

- One of the most widely used clustering algorithms.
- Objective: divide data into K groups (clusters) that minimize within-cluster variability (i.e., the within-cluster distances).
- Each observation is assigned to the cluster with the *nearest* mean (centroid).

Example:

- Suppose we have county-level data on obesity rate, smoking rate, and median income.
- K-means can group counties into clusters that share similar health risk profiles.

K-means: Objective Function

- K-means minimizes the total within-cluster sum of squares:

$$\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

  where μ_k is the mean of cluster C_k.
- Algorithm steps:
  1. Choose K (number of clusters).
  2. Randomly assign points to clusters.
  3. Compute cluster means.
  4. Reassign points to the nearest mean.
  5. Repeat until assignments stop changing.

Choosing K

- For all of the clustering algorithms, we need to specify K, the number of clusters in the given dataset.
- Common methods:
  - **Elbow method:** plot total within-cluster sum of squares vs K.
  - **Gap Statistic:** measures when the improvement in fit is not better than what is expected by chance.
- In public health, interpretability also matters—choose K that makes sense in context (e.g., 3-5 community profiles).

Elbow method

- Let W_k be equal to the within-cluster distances for a given algorithm when the number of clusters is k.
- Let k^* be the true number of clusters.
- We'd expect that

$$\{W_k - W_{k+1} | k < k^*\} \gg \{W_k - W_{k+1} | k \geq k^*\}$$

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  - The difference in consecutive W_k values will be larger when $k < k^*$.
- An estimate of K is then obtained by identifying a "kink" in the plot of W_K as a function of K.

Gap Statistic

- A method to determine if $W_k - W_{k+1}$ is "large" or "small" is the Gap Statistic.
- This is a Monte Carlo method, multiple datasets with no clusters are generated with a similar spread to the variables in the original data.
- **Random Gap Values:** the $W_k - W_{k+1}$ values from these *random* datasets
- Use the Random Gap Values to judge if the observed values are larger than what would be expected by chance.

Gaussian Mixture Models (GMMs)

- K-means assumes spherical clusters of similar size.
- Gaussian Mixture Models (GMMs) assume each cluster follows a multivariate normal distribution.
- Each observation has a **probability of belonging** to each cluster.

Example:

- In a nutrition dataset, GMMs can identify overlapping groups such as "mostly healthy eaters" vs "moderate" vs "unhealthy."

Expectation-Maximization (EM) Algorithm

- Used to estimate GMM parameters.
- Alternates between:
  - **E-step:** Estimate cluster membership probabilities.
  - **M-step:** Update cluster means and covariances based on probabilities.
- Stops when parameter changes are very small.
- Output includes both the most likely cluster assignment and the uncertainty of membership.

Partitioning Around Medoids (PAM)

- Similar to K-means but uses **medoids** (actual observations) instead of means.
- More robust to outliers because medoids minimize total distance within clusters.
- Works with any distance measure (not just Euclidean).

Example:

- PAM could be used to cluster hospitals using a mix of numeric and categorical performance indicators.

Hierarchical Clustering

- Hierarchical Clustering is a procedure that results in different degrees of clustering.
- There are two ways to do this:
  1. **Agglomerative:** Bottom-up: each point starts as its own cluster and ends up in one cluster.
  2. **Divisive:** Top-down, everyone starts as one cluster and ends up in their own cluster.
- These have similarities to forward-backward selection methods.
- For either method, observations are joined (or separated) by how dissimilar they are.

Hierarchical Clustering

Algorithm for agglomerative clustering:

1. Everyone starts in their own cluster.
2. Find the smallest d_{ij}, say d_{IJ}. Put I and J in the same cluster.

Now we have to recompute the dissimilarities for the new group IJ.

3. Compute dissimiarities $d_{IJ,k}$ for all $k \neq I$ or J. Three options for doing this:
   - **single**: $d_{IJ,k} = \min(d_{Ik}, d_{Jk})$
   - **complete**: $d_{IJ,k} = \max(d_{Ik}, d_{Jk})$
   - **average**: $d_{IJ,k} = avg(d_{Ik}, d_{Jk})$
4. Repeat steps 2–3 $n - 1$ times.

Hierarchical Clustering

- Single is more likely to cluster an observation with a group of observations.
- Complete is more likely to create new clusters.

- The algorithm for divisive clustering begins with the entire data set as a single cluster, and recursively divides one of the existing clusters into two daughter clusters at each iteration in a top-down fashion.
- We won't discuss how this works, and divisive clustering is less common than agglomerative.

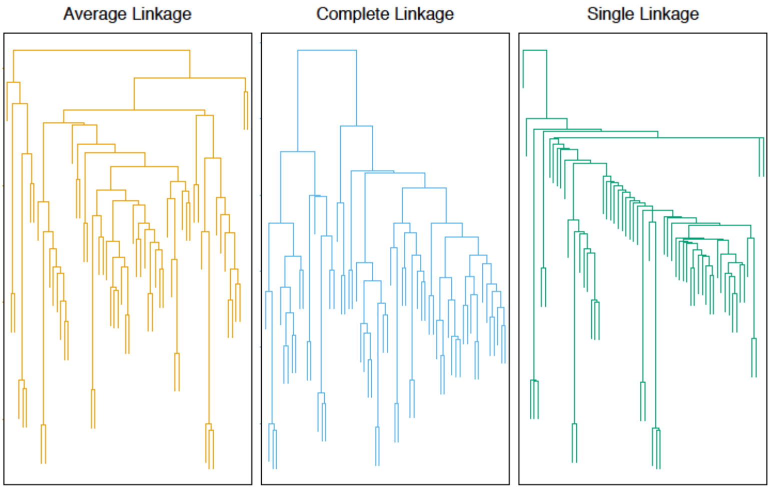

Figure: From page 524 in the online version of ESL.

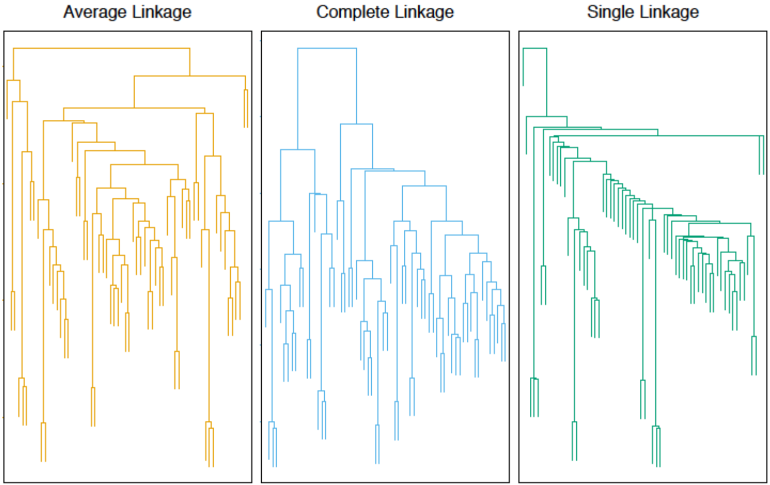

Figure: From page 524 in the online version of ESL.

Using Clustering in Downstream Models

- In practice, clustering results are often used as predictors or group indicators in later analyses.
- **Examples:**
  - Use cluster membership as a covariate in a regression model for predicting disease risk.
  - Compare outcomes (e.g., mortality, quality of life) across clusters identified from biomarker or behavior data.
  - Use clusters to define subgroups for targeted interventions or policy evaluation.
- This approach allows us to connect unsupervised patterns with meaningful outcomes.
- **However,** it changes the scientific question: we are no longer just describing structure, but testing hypotheses *about the clusters themselves*.

Pitfalls and Recommendations

- **Main Concern:** The same data are used to *create* and *test* clusters.
- This can lead to:
  - Overstated associations (data snooping bias)
  - Invalid p-values or confidence intervals
  - Misleading conclusions about "differences" between clusters
- **Good Practices:**
  1. If possible, use an independent dataset to test relationships between clusters and outcomes.
  2. Use *cross-validation* or bootstrapping to assess the stability of the clusters.
  3. Treat clustering results as *exploratory*—use them to generate, not confirm, hypotheses.
  4. Report how sensitive your downstream results are to the chosen number of clusters and method.

When to Use Each Method

- **K-means:** fast, works well for continuous data with spherical clusters.
- **GMM (model-based):** captures overlapping and elliptical clusters.
- **PAM:** robust to outliers, works with mixed data types.
- **Hierarchical:** good for exploring structure or visualizing relationships.

Key takeaway: Always pair clustering with interpretation and domain expertise—machine learning finds structure, but you decide if it makes sense.

Summary

- Clustering is essential for exploratory analysis in public health and many other fields.
- Always pre-process data: scale variables, handle outliers, and ensure numeric format.
- Use visualization to interpret results (e.g., scatter plots, silhouette plots, dendrograms).
- Combine statistical performance and interpretability when deciding the number of clusters.
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